

Q1. (a) How do tubular batteries differ from other types of batteries?

(b) How can I calculate the backup time of an inverter with specific load?

Prathesh M.
Through e-mail

A1. (a) Basically, there are two types of batteries: flat pasted plate and tubular. These batteries get their name from the design of the positive plate.

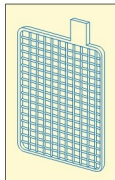


Fig. 1: Flat-pasted plate design

The negative plates are identical in both types. One major difference is that in the flat-plate design the positive plate is a rugged lead-alloy grid which is filled with a specially compounded paste of active material,

whereas in the tubular design the positive plate is composed of a series of parallel tubes filled with lead oxide. **Flat-pasted plate.** The plates are made with a rugged cast grid usually made from a lead alloy containing 3-6 per cent antimony. A pure-lead grid structure is not strong enough by itself to stand vertically while supporting the active material. Other metals in small quantities are alloyed with lead for added strength and improved electrical properties. Most commonly alloyed metals are antimony, calcium, tin and selenium.

The grids are pasted with a specially compounded mixture of lead-oxide (finely divided lead) water and sulphuric acid. The plates are 'cured' by a process that converts the active material in the plate into the desired composition, causing the paste to set to a hard cement.

Tubular plate. The design of tubular-plate batteries is more complex than flat pasted plate. The grid is a series of parallel lead rods or spines cast onto a bar. It is usually fabricated from a lead alloy containing 6-10 per cent antimony. Series of porous glass-fibre

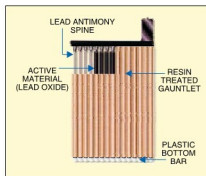


Fig. 2: Tubular plate design

tubes are fitted in parallel over the grid spines. These tubes are then filled with a mixture of lead oxide and red-lead powder. The filled tubes are sealed with end caps. The resulting assembly is then 'pickled' by soaking in dilute sulphuric acid to convert the lead oxide into lead sulfate. The finished product comprises a series of tubes filled with lead sulphate and a centre core of lead to carry the current.

Tubular batteries have higher capacity-to-size ratio. These batteries can be recharged faster and are engineered to deliver increased power and higher efficiency. These are widely used for telecommunication, emergency lighting, signaling and power devices.

(b) The backup time is simply the number of hours for which an inverter will be able to run the lighting load during power failure.

The backup time in hours = $Ah \times 12V \times N \times PF \times EFF / \text{Load (in VA)}$ where 'Ah' is the ampere-hour capacity of the battery, 'N' stands for the number of 12V batteries in use, 'PF' stands for the power factor of the inverter, 'EFF' stands for the efficiency (0.9) of the battery and 'load' is the sum of VA ratings of the tubes and fans connected to the inverter.

Q2. Some circuits use specific ratings of relays. I want to know:

- How does a relay operate?
- How to use it in a circuit?
- How to connect its five pins?

Regan
Chennai

A2. Relays are electromechanical/solidstate devices used to iso-

late one electrical circuit from another. These allow a low-current control circuit to make or break an electrically isolated, high-current circuit path. Systems based on the use of a relay have a controlling block that gets a low voltage—the controlling signal—for example, from a 555 timer or microcontroller, to control

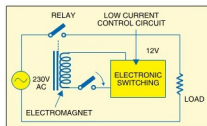


Fig. 3: Relay with controlling block

another block which is isolated and operating on a high voltage.

Relays are used to:

- Control a high-voltage circuit with a low-voltage signal
- Control a high-current circuit with a low-current signal, as in the starter solenoid of an automobile
- Detect and isolate faults on transmission and distribution lines by opening and closing circuit breakers (protection relays)

The information normally available on the casing of a relay includes current rating of contacts in amperes, alternating or direct current, voltage rating of contacts, coil voltage (usually 5V, 6V, 12V) and layout. Coil is connected to the control circuit. Line terminal of the 230V AC supply is connected to pole. Load is connected between 'N/O' and neutral. When the coil is energised, pole is connected to 'N/O' and line is extended to the load to complete the circuit.

Answers compiled by Col (Retd) N.C. Pandey

Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase I, New Delhi 110020 (E-mail: editsec@efyindia.com) and should include name and address of the sender.